1. Explain Newton's Laws of Motion with examples.

Newton's First Law: An object at rest stays at rest, and an object in motion stays in motion unless acted upon by an external unbalanced force. For example, if you kick a stationary ball, it starts moving because of the applied force.
If you stop pushing a moving cart, it eventually comes to rest due to friction slowing it down.

Newton's Second Law: Force equals mass times acceleration. ($F = ma$). This means a greater force makes an object accelerate more. For instance, pushing a heavy box requires more force than pushing a light one to achieve the same acceleration.

Newton's Third Law: For every action, there is an equal and opposite reaction. If you push on a wall, the wall pushes you back with the same force. When a rocket launches, gases expelled downwards push the rocket upwards.

2. Discuss the water cycle in detail.

The water cycle describes how water moves from the Earth's surface to the atmosphere and back.

First, water from oceans, lakes, and rivers evaporates due to the sun's heat. This water vapor rises into the atmosphere.

In the atmosphere, the water vapor cools and condenses into tiny water droplets, forming clouds.

When these droplets combine, they become heavy and fall back to the Earth's surface as rain, snow, sleet, or hail.

This water then collects in bodies of water like rivers, lakes, and oceans,